IRL Questions

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- **Question 1**
  - Recall the definition of the generalized Mittag–Leffler function

    $$E_{\alpha,\beta}(z) = \sum_{k=0}^{\infty} \frac{z^k}{\Gamma(\alpha k + \beta)}.$$

    Show that for $\alpha = 1, \beta = 1$, we recover the exponential function $E_{1,1}(z) = e^z$.

- Recall the **integral definition** (for $\Re(r) > 0$):

  $$\Gamma(r) = \int_0^{\infty} u^{r-1} e^{-u}\, du.$$

- Use the **recursion**:

  $$\Gamma(r+1) = r\,\Gamma(r).$$

- Apply the Taylor's theorem formally and then apply it to the exponential function.

- **Question 2**
  - Using the definition above, prove that

    $$E_{1,2}(x) = \frac{e^x - 1}{x}, \quad x \neq 0.$$

    Hint: separate the $k = 0$ term and compare with the Taylor expansion of e^x.

- **Question 3**
  - Compute by hand the integral

    $$\int_1^2 \frac{1}{x} dx$$